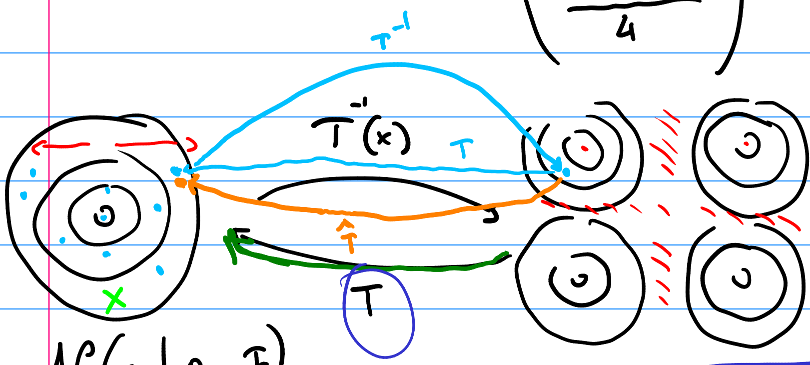
Let $\overset{\text{Exact map}}{T(x)} = \nabla\left(\frac{x^4+y^4}{4}\right) = \begin{pmatrix} x^3 \\ y^3 \end{pmatrix}$

$$T^{-1}(x) = \begin{pmatrix} x^{\frac{1}{3}} \\ y^{\frac{1}{3}} \end{pmatrix}$$



$$\mathcal{N}(x \mid 0, I) = \frac{e^{-\frac{\bar{x}^T \bar{x}}{2}}}{2\pi} = \frac{e^{-\frac{\|\bar{x}\|^2}{2}}}{2\pi} = \frac{e^{-\frac{(x^2+y^2)}{2}}}{2\pi}$$

1. Sample $\mathcal{N}(0, I) \to \{(x_i, y_i)\}$
2. Use $T^{-1}(x_i, y_i) = (u_i, v_i)$
3. Given $S_1 = \{x_i, y_i\}$, $S_2 = \{(u_i, v_i)\}$. Comput $\hat{T}$ mapping S_2 into S_1 using the flow.
4. Check the accuracy of your numerically computed map $\hat{T}$

Ideally

$$T(u_i, v_i) = (x_i, y_i)$$

$$\text{ERROR} = \frac{\sum_{i=1}^{N} (\hat{T}_i^1 - x_i)^2 + (\hat{T}_i^2 - y_i)^2}{N}$$

where $\hat{T}(u_i, v_i) = \begin{pmatrix} \hat{T}_i^1 \\ \hat{T}_i^2 \end{pmatrix}$

⚠ This way of measuring the ERROR may be missleading when considering overfitting.

It is better to compute the error on a set of grid points (as opposed to compute it on S_1 and S_2)

Once you have your $\hat{T}^{-1}$

$$\boxed{E_g = \frac{1}{N_g}\sum_{j=1}^{N_g} \left(\hat{T}_i^1 - x_i^g\right)^2 + \left(\hat{T}_i^2 - y_i^g\right)^2}$$

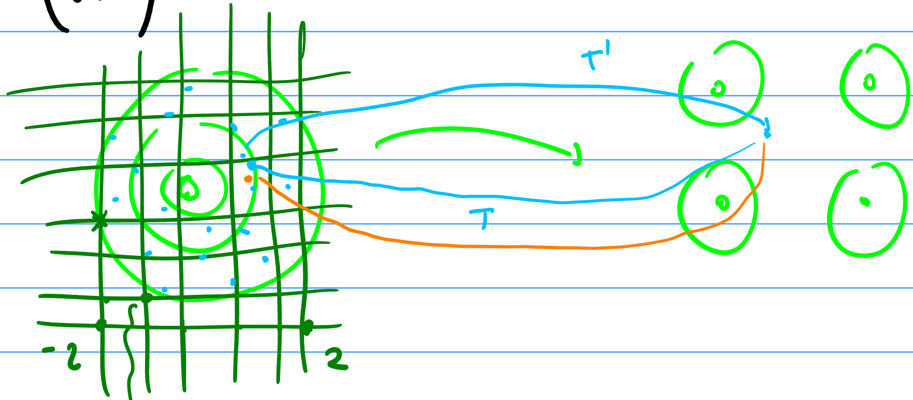
↓ Error on a grid.

where $\hat{T}_i^1$ now is equal to

$$\hat{T}_i^1 = \hat{T}(u_i^g, v_i^g)$$

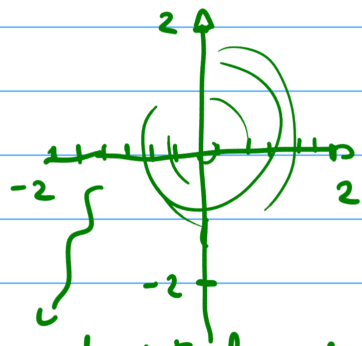
$$\hat{T}_i^2 = \hat{T}(u_i^g, v_i^g)$$

$$\begin{pmatrix} u_i^g \\ v_i^g \end{pmatrix} = T^{-1}(x_i^g, y_i^g)$$



(x_i^g, y_i^g)

take $[-2, 2] \times [-2, 2]$



100 grid point for each dimension

$\Rightarrow N_g = 100 \times 100 = 10^4$

⚠ You can plot directly $T(x_i^g, y_i^g)$ & $\hat{T}(x_i^g, y_i^g)$ (These are 2D plots)